
EECS 16B Designing Information Devices and Systems II

Discussion Worksheet Discussion 2B

For this discussion, **Note 1** is helpful.

1. RC Circuits

In this problem, we will be using differential equations to find the voltage across a capacitor over time in an RC circuit. We set up our problem by first defining four functions over time: $I(t)$ is the current at time t , $V(t)$ is the voltage across the circuit at time t , $V_R(t)$ is the voltage across the resistor at time t , and $V_C(t)$ is the voltage across the capacitor at time t .

Recall from 16A that the voltage across a resistor is defined as $V_R = RI_R$ where I_R is the current across the resistor. Also, recall that the voltage across a capacitor is defined as $V_C = \frac{Q}{C}$ where Q is the charge across the capacitor.

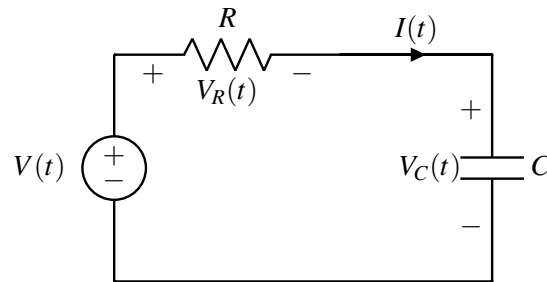


Figure 1: Example Circuit

- (a) First, find an equation that relates the current across the capacitor $I(t)$ with the voltage across the capacitor $V_C(t)$.

(b) Write a system of equations that relates the functions $I(t)$, $V_C(t)$, and $V(t)$.

(c) So far, we have three unknown functions and only one equation, but we can remove $I(t)$ from the equation using what we found in part (a). Rewrite the previous equation in part (b) in the form of a differential equation.

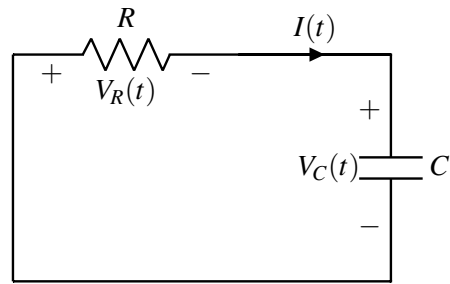


Figure 2: Circuit for part (d)

- (d) Let's suppose that at $t = 0$, the capacitor is charged to a voltage V_{DD} ($V_C(0) = V_{DD}$). Let's also assume that $V(t) = 0$ for all $t \geq 0$. Solve the differential equation for $V_C(t)$ for $t \geq 0$.

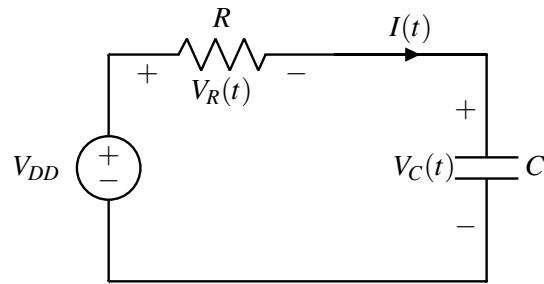


Figure 3: Circuit for part (e)

- (e) Now, let's suppose that we start with an uncharged capacitor $V_C(0) = 0$. We apply some constant voltage $V(t) = V_{DD}$ across the circuit. Solve the differential equation for $V_C(t)$ for $t \geq 0$.